SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

   Constraints:

   i. D1 cannot work Night shift
   ii. D2 must work before D3 (Morning < Afternoon < Night)
   iii. Only one doctor per shift
   iv. D3 cannot work Morning
   v. All doctors must be assigned exactly one shift

   Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

   Constraints:

   i. Movement allowed: Up, Down, Left, Right
   ii. Cost of each move = 1
   iii. Cannot pass through obstacles
   iv. Use Manhattan Distance heuristic

   Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

a) Analyze all the given constraints and explain how they affect the robot’s decision-making process.

b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.

c) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents
- Rationality
- Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Question 1

Problem Understanding

This is a **Constraint Satisfaction Problem (CSP)** where three doctors (D1, D2, D3) must be assigned to three different shifts. The objective is to satisfy all the given constraints using the **Backtracking Search** algorithm.

Constraint Analysis

The constraints ensure that each doctor is assigned exactly one shift without conflicts. They also specify restrictions such as D1 cannot work Night, D3 cannot work Morning, and D2 must be scheduled before D3.

Solution Development
Initial Domains

Doctor	**Possible Shifts**
D1	Morning, Afternoon
D2	Morning, Afternoon, Night
D3	Afternoon, Night

Step 1: Assign **D1 = Morning**
Step 2: Try **D2 = Morning** (Already occupied) → **Backtrack**
Step 3: Assign **D2 = Afternoon**
Step 4: Try **D3 = Afternoon** (Already occupied) → **Backtrack**
Step 5: Assign **D3 = Night**
All constraints are satisfied.

Search Tree

```
            Start
              |
        D1 = Morning
           /     \
   D2=Morning     D2=Afternoon
     (Fail)          |
              D3=Afternoon
                 (Fail)
                   |
               D3=Night
                Success
```

Final Schedule

Doctor	**Shift**
D1	Morning
D2	Afternoon
D3	Night

Question 2

Problem Understanding
The robot must move from the **Start (S)** position to the **Goal (G)** in a 5×5 grid. It can move only in four directions while avoiding obstacles and finding the shortest path using the **BFS**

Constraint Analysis
The robot can move only **Up, Down, Left, and Right**, and each move has a cost of **1**. Obstacles cannot be crossed, and the **Manhattan Distance heuristic** is used to estimate the distance to the goal.

Manhattan Distance Formula

$$h(n) = | x_1 - x_2 | + | y_1 - y_2 |$$

For the Start node:

$$h(S) = | 4 - 0 | + | 4 - 0 | = 8$$

BFS Queue Updates
Initial Queue : [S]
Expand S
Queue : [(0,1),(1,0)]
Expand (0,1)
Queue : [(1,0),(0,2)]
Expand (1,0)
Queue : [(0,2),(2,0)]

Continue until Goal (G) is reached.

S → (0,1) → (0,2) → (1,2)
→ (2,2) → (2,3) → (2,4)
→ (3,4) → G

Total Cost = 8

Application of AI Concepts
BFS is an **uninformed search algorithm** that explores nodes level by level. It guarantees the shortest path in an unweighted grid and avoids revisiting already explored nodes.

Justification
BFS is suitable because every movement has the same cost. It systematically explores all possible paths and always finds the shortest path from the Start to the Goal.

Conclusion
The robot reaches the goal through the shortest path with a **total cost of 8**. Hence, BFS successfully solves the navigation problem.

Question 3

Problem Understanding
An autonomous rescue robot must reach a survivor in a disaster-affected building. The robot operates in a **partially observable environment** and must find the safest path while minimizing the total travel cost.

Constraint Analysis
The robot can move only in four directions and cannot cross obstacles. It can sense only adjacent cells, risky zones increase the travel cost, and it must continuously update its knowledge while moving.

Solution Development

Chosen Algorithm: Uniform Cost Search (UCS) with an Online Search Agent
Steps

1. Start from **S**.
2. Sense adjacent cells.
3. Update the internal map.
4. Expand the node with the **lowest cumulative cost**.
5. Avoid blocked and risky paths whenever possible.
6. Repeat until the robot reaches **G**.

Flowchart

Start
  |
Sense Adjacent Cells
  |
Update Knowledge
  |
Expand Lowest Cost Node
  |
Goal Reached?
  |
No → Continue Search
  |
Yes
  |
Reach Survivor

Application of AI Concepts
The rescue robot acts as an **Intelligent Agent** by sensing the environment and taking appropriate actions. It follows **Rationality** by selecting the safest and lowest-cost path, and it operates in a **partially observable, dynamic, sequential, deterministic, and discrete environment**.

Conclusion The rescue robot successfully reaches the survivor by minimizing the total cost while avoiding obstacles and risky zones. Therefore, **Uniform Cost Search with an Online Search Agent** is the most appropriate solution for this problem.